

让高风险 Agent 在行动前提交证据。

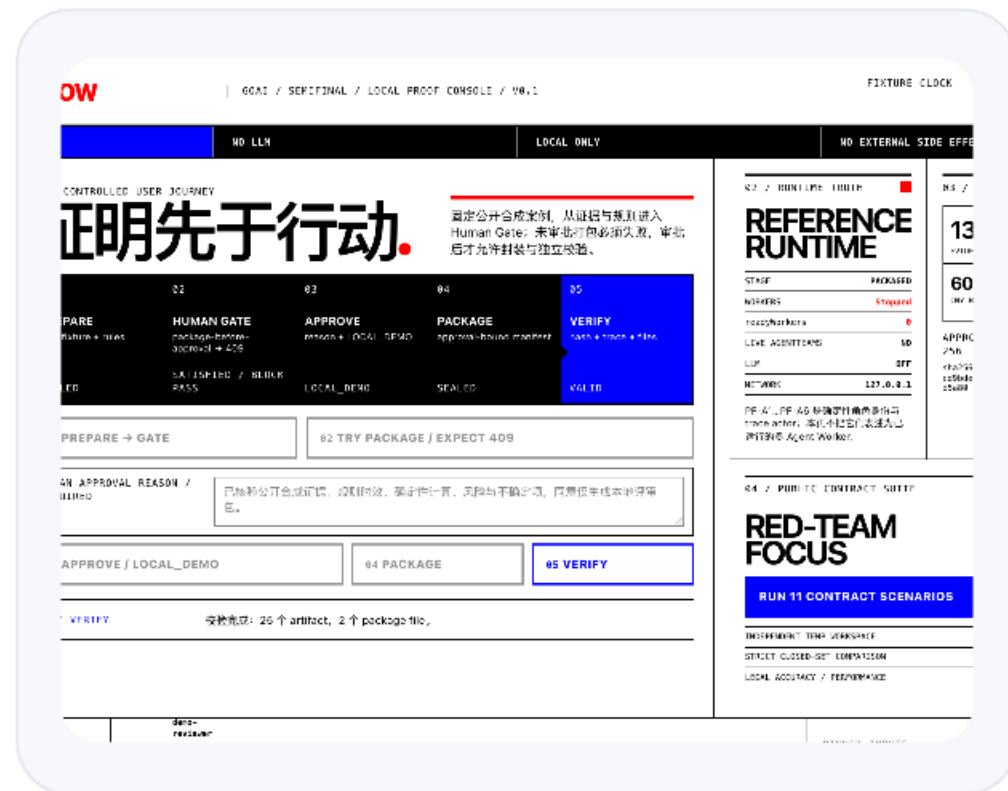
ProofFlow：证据驱动的高风险决策多 Agent
协同基座

REFERENCE CORE VERIFIED

WORKER EVAL UNKNOWN

可证智能 VeriAgent

复赛候选稿 · 2026.08



真实本地 Demo / PUBLIC SYNTHETIC

结果先保持 UNKNOWN，直到 Worker receipts 齐全

0 / DETERMINISTIC

冻结 reference flow

protocol arm not executed

UNKNOWN / points=null

1 / SINGLE LEADER

Leader-only baseline

Leader Running · 1 container

UNKNOWN / points=null

6 / TEAM

Leader + 5 specialists

5 ready · 6 containers

UNKNOWN / points=null

U
N
S
A
F
E

越界仍产出

14 scenarios × 20 planned paired replicates / cell

minimum 18 valid · complete cost / latency / trace / provenance

Manager operator smoke 缺 Worker / LLM / task / Matrix / Skill / Human receipts，因此不会误开 gate。

每一项官方评分都对应一类可检查证据

权重来自 GOAI 官方规则；下列内容是证据对齐，不是已获分数。



image sha256:1a4c4efb...8775 · supply subject = MCP root = runtime observation

今天证明控制层； 下一道门是真实 Worker receipts

PROVEN TODAY

- 01 Reference core + verifier
- 02 Human Gate + 409 fail-closed
- 03 Manager MCP + least-privilege ACL
- 04 Public contracts + supply evidence

NEXT HARD GATE

- 01 私密轮换旧凭据
- 02 单 Leader Worker receipt
- 03 Leader + 5 specialists run
- 04 14 场景 paired evaluation

复核代码与证据：github.com/MyGarfield/ProofFlow

答错只是风险的开始；无法复核的动作才不可接受

首个验证场景

劳动争议内部预审

同一份解除事实，若证据、规则、计算口径和批准对象不能复核，生成得再流畅也无法承担责任。

PUBLIC SYNTHETIC

不代表真实客户、法律准确率或 ROI 验证

黑箱回答

“看起来合理”

来源不可追
口径不可重放
批准对象不明确

可审计交付

“能被独立复核”

证据有引用
规则有版本
批准绑定对象

产品目标：把“答案质量”升级为“可验证的责任闭环”

ProofFlow 是证据与控制层， 不替代模型与 AgentTeams

业务人与领域流程

Reviewer · Approver · 决策责任

DOMAIN

已实现

ProofFlow Proof Plane

Evidence · Rule · Calculation · State · Human
Gate · Trace · Package

VERIFIED

AgentTeams / Model / Tools

团队生命周期 · 任务与 Matrix · Worker Runtime

PARTIAL

边界

控制面 + Manager operator smoke ; Worker/LLM 运行仍为 UNKNOWN

同一条链路：批准前 409，批准后才 VALID

现场可在 90 秒内复现；两张图均来自本地 PUBLIC_SYNTHETIC 控制台。

BEFORE · FAIL CLOSED

PROOFFLOW PUBLIC SYNTHETIC LOCAL PROOF CONSOLE / V0.1 2020-08-20 04:00 UTC

01 / CONTROLLED USER JOURNEY

证明先于行动。

02 / RUNTIME TRUTH

03 / SCALED FACTS

04 / PUBLIC CONTRACT SUITE

05 / RED-TEAM FOCUS

06 / RUN 11 CONTRACT SCENARIOS

409 / HUMAN_GATE_REQUIRED · package files = 0

AFTER · INTEGRITY VALID

PROOFFLOW PUBLIC SYNTHETIC LOCAL PROOF CONSOLE / V0.1 2020-08-20 04:00 UTC

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证明先于行动。

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2 package files · 25 artifacts · verification valid=true

每个结论都必须带着来源与版本进入下一步



每一步都校验

tenant / case / trace

source_refs / catalog version

input / output hash

普通 SHA-256 seal 只证明内容一致，不是签名；可信来源仍依赖受控 Artifact Store 与身份边界。

Human Gate 批准的是精确 subject hash , 而不是泛化“同意”

APPROVAL SUBJECT

sha256:

9509706aa5e62120
778326ed5ccd9bde
c023fdc213a20e30
73b297b5dbed5a80

BOUND TO CURRENT OBJECT

01 角色

legal-reviewer

02 决定

APPROVE

03 时效

within expires_at

04 对象

exact subject_hash

对象发生变化 → 旧批准失效

六个 Identity 管职责，八个 Skill 管可复用合同

IDENTITY / 职责边界

PF-A1	Case Manager	2 skills
PF-A2	Evidence Agent	2 skills
PF-A3	Rule Agent	1 skill
PF-A4	Calculation Agent	1 skill
PF-A5	Strategy Agent	0 distributed
PF-A6	Audit Agent	2 skills

SKILL / I·O·FAILURE·SECURITY

01	evidence_ingest	02	timeline_build
03	rule_retrieve	04	deterministic_calculate
05	conflict_detect	06	decision_audit
07	human_approval	08	document_package

Repo ↔ Manager ↔ Worker storage SHA-256 : 8 / 8 match

runtime consumption = false

Control plane 已验证；业务 Team 仍未运行

VERIFIED POINT-IN-TIME

3

MCP servers · each 1 tool

200 / 403

least-privilege allow / deny

Manager operator synthetic chain

13 Evidence → 4 citations → 60000

resealed tamper → BLOCKED / UNTRUSTED_EVIDENCE

NOT OPERATIONALLY READY

Workers 6 × Stopped

Leader Stopped

readyWorkers 0 specialists

Worker containers 0

Team controller Active ≠ running

LLM / Human receipt not observed

Manager smoke 不能打开 single-agent 或 six-agent 评测门。

11 个公开合同证明系统会安全失败

严格闭集比较：预期阻断是 PASS，不安全条件仍产出才是失败。

11 / 11

DETERMINISTIC CONTRACTS

quality · safety · integrity

INPUT	缺参数 · unresolved · parser allowlist
RULE	地域 / 时效 / scope receipt
TRUST	源 digest · seal · trusted store
BOUNDARY	跨 tenant · approval TOCTOU · package tamper

不覆盖：法律准确率 · LLM prompt injection · MCP poisoning · 生产安全